

DATA VORTEX 2026

ROUND 4 — ANALYTICAL FINDINGS REPORT

Empirical Findings Across Forensic Recovery, Relational Aggregation, NLP, and Social Signals

September 2026 | Final Analytical Synthesis

1. Cross-Stage Synthesis & Evidence Matrix

Across four competitive stages, the Social Engine was forensically recovered, relationally analyzed, semantically classified, and monitored for public reaction to algorithmic distribution. All numbers reported below are traced directly to verified canonical deliverables.

Analytical Metric	Canonical Value	Significance / Interpretation
Cleaned Posts Volume	12,000	Exactly 1,000 duplicate posts safely removed; zero data loss.
Cleaned Creators	1,500	Relational foreign-key integrity validated across 100% of posts.
Total Verified Likes	1,250,919	Negative-like values restored without discarding legitimate records.
NLP Sentiment Baseline	0.5795 Macro F1	TF-IDF + Logistic Regression on 1,794 held-out test posts.
NLP Topic Accuracy	0.9264 Accuracy	TF-IDF + Linear SVM across 4 domain topic categories.
HN Archive Sample	204 records	Public reaction records across 15 years (2011–2026).
Negative Reaction Skew	64.71%	Strong public skepticism towards automated recommendation changes.
Candidate Shift SS1	Delta -0.80	Descriptive drop from 2025-09 (index -0.20) to 2025-10 (-1.00).
Candidate Shift SS2	Delta +0.80	Descriptive rebound from 2024-10 (index -1.00) to 2024-11 (-0.20).
Engagement Spike ES1	1,704 points	March 2023 spike coinciding with Twitter algorithm open-sourcing.

2. Round 1 Analytical Insights: Skew & Location Leadership

SQL challenge E2 identifies extreme engagement concentration in top content items. Challenge M1 confirms that user locations vary significantly in collective interaction volume, while challenge H2 reveals local creator leadership partitioned by geographical location.

3. Round 2 NLP Insights: Balanced Polarity vs. Thematic Imbalance

Supervised training on 9,000 posts demonstrated that while sentiment classes were balanced (3,000 each), topical categories were heavily dominated by Community Discussion (7,752 posts). GroupShuffleSplit prevented 987 duplicate-normalized text groups from crossing partitions, ensuring honest generalization bounds.

4. Round 3 Signal Insights: Algorithmic Reactions

The 204 retained public records from Hacker News demonstrate a prevailing negative sentiment (132 Negative, 38 Neutral, 34 Positive). Entity extraction confirmed YouTube (70 mentions), Twitter (66 mentions), and Facebook (59 mentions) as the dominant platforms associated with feed and algorithm critique.

5. Methodology & Explicit Limitations

The following boundaries are documented without exaggeration:

- **Source Demographics:** Hacker News represents an engineering-focused community rather than a population-representative social sample.
- **Historical Archive:** The Round 3 dataset is a bounded historical archive sample, not a live streaming feed.
- **Causal Boundaries:** All detected shifts (SS1, SS2) and engagement spikes (ES1, ES2, ES3) are empirical temporal associations. Causality is strictly disclaimed.
- **Missing Values:** Historical comment scores are missing by API design and are excluded rather than zero-imputed.